

Math Bootcamp Syllabus

For Incoming MSc Economics Students

Math Camp 2026

Overview

- **Objective.** Bridge a weaker undergraduate math background to MSc-level economics (micro, macro, econometrics, mathematical economics).
- **Format.** 5 days $\times$ 3 hours. Emphasis on intuition and standard problem templates; only simple proofs where needed. Detailed proofs live in slide appendices / fall courses.
- **Materials.** Day-by-day lecture slides; this syllabus is the concise roadmap.

Day-by-day topics

Day 1 — Foundations (§1.1–1.9)

1. **Notation, logic, sets, functions.** Number systems; set membership and logic symbols; necessary/sufficient/contrapositive; set operations and Cartesian products; domain/codomain/range; injective/surjective.
2. **Sup/inf and algebra.** max / min vs. sup / inf; logs, elasticities, Cobb–Douglas; homogeneity and Euler’s theorem (CRS adding-up).
3. **Calculus.** Single-variable derivatives (rules, marginal interpretation); multivariate: partials, total differential, gradient, directional derivatives, cross-partial; limits, continuity, differentiability.

Day 2 — More calculus and analysis bridge (§2.1–2.6)

1. **Implicit function theorem.** Comparative statics when $y^*(\theta)$ is only defined implicitly; tax incidence.
2. **Existence tools and integration.** Open/closed/compact sets; EVT / Weierstrass; MVT; Riemann integral and FTC; expectation via CDF (integration by parts).
3. **Leibniz, Fubini, Taylor, uniform convergence.** Differentiating under the integral; changing order of integration; linearization / Taylor; why uniform convergence matters for exchanging limits.

Day 3 — Linear algebra and real analysis

1. **Linear algebra.** Vectors, matrices, $Ax = b$, transpose, determinants (2×2), invertibility, linear independence, rank.
2. **Real analysis bridge.** Uniform-convergence practice; separation / supporting hyperplanes (welfare intuition); fixed-point existence statements (Brouwer / contraction, no proofs).

Day 4 — Optimization (§4.1–4.5)

1. **Unconstrained optima.** FOC/SOC; concavity via f'' / Hessian; local vs. global.
2. **Shape and equality constraints.** Concave vs. quasiconcave; convex upper contour sets; Lagrange multipliers; shadow prices; envelope theorem.
3. **Inequalities and KKT.** Binding vs. slack; complementary slackness; geometric Farkas $\Rightarrow$ KKT; corner solutions (nonnegativity).

Day 5 — Dynamic programming

1. **Framework.** Sequential decisions; state vs. control; law of motion; finite-horizon focus and Markov idea.
2. **Bellman and solution method.** Value function and principle of optimality; Bellman equation; backward induction; policy function.
3. **Worked core.** Two-period consumption–savings; Euler equation; link to standard intertemporal optimization; VFI intuition (infinite horizon, conceptual only).

Cross-cutting goals

- Read FOCs, SOC, and comparative statics fluently.
- Map math objects to economic meaning (marginals, shadow prices, value functions).
- Build confidence with the templates used in fall core courses.

Suggested time allocation

Day	Theme	Block 1 (~1 hr)	Block 2 (~1 hr)	Block 3 (~1 hr)
1	Foundations	Notation, logic, sets, functions	Sup/inf; algebra; Cobb–Douglas	Derivatives; Euler; multivariate; limits
2	Calculus bridge	IFT	EVT/MVT; integration	Leibniz/Fubini; Taylor; uniform conv.
3	Lin. alg. / analysis	Linear algebra	Convergence practice	Separation; fixed points
4	Optimization	FOC/SOC; quasiconcavity	Lagrange; envelope	Farkas geometry; KKT
5	Dyn. prog.	Framework; state & control	Bellman equation	Backward induction; VFI idea

Note. Appendices on the slides (rigorous proofs, extra examples) are for interest / self-study and are not required classroom coverage.